

# Endurance of Unmanned Helicopters

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## 1 Introduction

Using basic physical arguments, we derive a parameterized formula for the time that an unmanned helicopter can remain aloft in a stationary hover (i.e., fixed vertical and horizontal positions) given a finite supply of fuel. Our model takes into account the effects of wind as well as the mass and power consumption of a possible payload. As a point of comparison, we adopt the parameters (mass, rotor diameter, etc.) of the A-160 Hummingbird and verify that its quoted endurance agrees roughly with our estimates. We conclude with a theoretical scaling study based on the assumption of geometric similarity.

## 2 Thrust

A helicopter generates vertical thrust by pulling surrounding air through the circular area swept out by the rotor blades. Suppose that the rotor disk has area  $A_r$  and that the speed of the air through the disk is  $u_r$  (see Fig. 1). Momentum conservation implies that the net thrust  $T$  provided by the rotor is of order the rate at which momentum in the air crosses the disk. The momentum flow rate is  $\rho_a u_r^2 A_r$ , where

$$\rho_a \approx 1.2 e^{-z/8} \text{ kg m}^{-3} . \quad (1)$$

is the density of air at altitude  $z$  (in km) above sea level. Because of drag on various components of the helicopter fuselage, the thrust is generally less than the total momentum flow rate. We take these effects into account by writing

$$T = \eta_t \rho_a u_r^2 A_r , \quad (2)$$

where  $\eta_t \lesssim 1$  is a dimensionless loss parameter.

When the rotation axis of the rotor is tilted away from the vertical, the total thrust vector has a component parallel to the ground, allowing the helicopter to move horizontally. If the rotor axis is tilted by an angle  $\theta$  to the vertical (see Fig. 1), the vertical and horizontal components of the thrust are  $T_z = T \cos \theta$  and  $T_x = T \sin \theta$ , respectively. In order for a helicopter of total mass  $M$  to maintain a fixed hovering altitude, the vertical thrust must balance the force of gravity, so that

$$T_z = \eta_t \rho_a u_r^2 A_r \cos \theta = Mg . \quad (3)$$

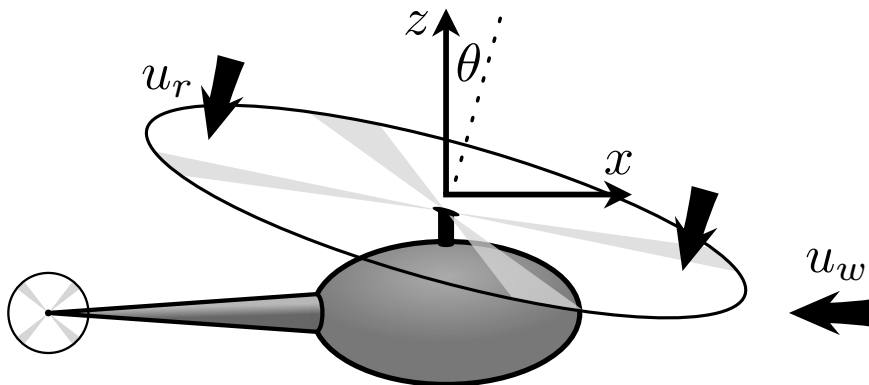


Figure 1: A helicopter of total mass  $M$  with rotor area  $A_r$ .

Given  $\theta$ , this equation allows us to compute  $u_r$ . We can determine  $\theta$  if we can specify an independent relation for  $T_x$ .

If the helicopter is to remain fixed in space, then, just as  $T_z$  balances the force of gravity,  $T_x$  must balance the drag force exerted on the helicopter by a horizontal wind. We assume that the helicopter is oriented such that its nose and streamlined fuselage point into the wind so as to reduce drag as much as possible. For a wind speed  $u_w$ , horizontal force balance demands

$$T_x = \eta_t \rho_a u_r^2 A_r \sin \theta = \rho_a u_w^2 A_d . \quad (4)$$

Here  $A_d$  is a cross sectional area that gives the correct drag force. Contributions to  $A_d$  include the projected area of the fuselage, weighted by the effects of streamlining, as well as parasitic elements, such as the main rotor shaft, tail rotor, and payload components. Taking the ratio of eqs. (3) and (4), we find

$$\tan \theta = \frac{\rho_a u_w^2 A_d}{Mg} . \quad (5)$$

A simple scaling argument allows us to develop a rough quantitative expectation for the value of  $\theta$ . Begin by writing the total mass as  $M = \bar{\rho} A_d L$ , where  $\bar{\rho}$  and  $L$  are, respectively, the mean density and some characteristic length of the helicopter fuselage. Because of the combination of metallic and composite materials, and allowing for some void space in the helicopter, we expect that  $\bar{\rho} \sim 10^3 \text{ kg m}^{-3}$ . We now estimate that

$$\tan \theta = \frac{\rho_a u_w^2}{\bar{\rho} L g} \approx 0.01 \left( \frac{\rho_a / \bar{\rho}}{10^{-3}} \right) \left( \frac{u_w}{10 \text{ m s}^{-1}} \right)^2 \left( \frac{1 \text{ m}}{L} \right) . \quad (6)$$

Wind speeds are likely to be less than  $30 \text{ m s}^{-1}$  (60 knots). While eq. (6) hides many details, it does suggest that, for most helicopter designs with  $L \gtrsim 1 \text{ m}$ , it is probably quite reasonable to use the approximations  $\tan \theta \approx \sin \theta \approx \theta$  and  $\cos \theta \approx 1$ . The validity of this approximation should be reevaluated for small, light-density helicopters.

The small  $\theta$  approximation, where terms of order  $\theta^2$  and smaller are dropped in any Taylor expansions, allows us to solve eq. (3) for  $u_r$ :

$$u_r = \left( \frac{Mg}{\eta_t \rho_a A_r} \right)^{1/2} . \quad (7)$$

As a concrete example, if  $M = 10^3 \text{ kg}$  and  $A_r = 100 \text{ m}^2$  (scales appropriate for the A-160 Hummingbird; see § 4), then  $u_r \approx 10 \eta_t^{-1/2} \text{ m s}^{-1}$ . Note that in the small  $\theta$  limit, the flow speed is insensitive to the head wind.

### 3 Power

Air acquires both momentum and kinetic energy as it is pulled through the rotor disk. The combined effects of drag and the power draw from mechanical and electrical systems not associated with the main rotor require an engine power that is larger than given by the energy flow rate at the rotor disk. Let the total power that needs to be supplied by the engine be  $P_{\text{tot}} = P_r + P_p$ , where  $P_r$  is power provided to the rotor shaft, and  $P_p$  includes the power diverted to payload and other systems. In all that follows, we will assume that  $P_p$  is some constant average power and that  $P_p/P_r$  is typically  $\ll 1$ .

Because of drag on the rotor blades and other frictional contributions, we expect that  $Tu_r$ —the characteristic rate of mechanical work—is a fraction  $\eta_s \lesssim 1$  of the shaft power, and thus

$$P_r = \eta_s^{-1} \rho_a u_r^3 A_r . \quad (8)$$

Substitution of eq. (7) for  $u_r$  yields

$$P_r = (\eta_s \eta_t^{3/2})^{-1} \left( \frac{M^3 g^3}{\rho_a A_r} \right)^{1/2} . \quad (9)$$

Using the example parameters  $M = 10^3 \text{ kg}$  and  $A_r = 100 \text{ m}^2$ , we estimate  $P_r \approx 10^5 (\eta_s \eta_t^{3/2})^{-1} \text{ W}$ . As  $M$  decreases, so do the vertical thrust and shaft power needed to keep the helicopter hovering at a fixed altitude.

An internal combustion engine provides power by converting the chemical energy of fuel into mechanical and electrical energy. For gasoline and diesel fuel, the available chemical energy per kilogram is  $\varepsilon_f \approx 50 \text{ MJ kg}^{-1}$ . If the fuel is burned at a rate  $\dot{M}_f > 0 \text{ (kg s}^{-1}\text{)}$  and a fraction  $\eta_f$  of the total chemical power is converted to useful power for the helicopter, then

$$P_{\text{tot}} = \eta_f \varepsilon_f \dot{M}_f . \quad (10)$$

Noting that  $\dot{M} = -\dot{M}_f$  and substituting eq. (9) for the shaft power, we obtain a differential equation for the total mass,

$$-\eta_f \varepsilon_f \dot{M} = (\eta_s \eta_t^{3/2})^{-1} \left( \frac{M^3 g^3}{\rho_a A_r} \right)^{1/2} + P_p . \quad (11)$$

It is convenient to scale the mass by the initial total mass  $M_0$  and change to the dependent variable  $\mu(t) = M(t)/M_0 < 1$ , with the initial condition  $\mu(0) = 1$ . The differential equation becomes

$$\dot{\mu} = - \left( \frac{1}{\tau_p} + \frac{1}{\tau_r} \mu^{3/2} \right) , \quad (12)$$

where

$$\tau_r = \frac{\eta_f \varepsilon_f M_0}{P_r(M_0)} = \eta_f \eta_s \eta_t^{3/2} \varepsilon_f \left( \frac{\rho_a A_r}{g^3 M_0} \right)^{1/2} \quad (13)$$

and

$$\tau_p = \frac{\eta_f \varepsilon_f M_0}{P_p} . \quad (14)$$

The constants  $\tau_r$  and  $\tau_p$  are the time scales for the rotor shaft and payload to completely consume a fuel mass equal to the total mass of the helicopter. When  $P_p/P_r$  is a small number,  $\tau_r$  is the shorter of the two timescales and is probably a good first approximation of the maximum flight time. Thus, an appropriate dimensionless time variable is  $\xi = t/\tau_r$ , which leads to the somewhat more appealing form of the differential equation,

$$\frac{d\mu}{d\xi} + \mu^{3/2} = -\alpha , \quad (15)$$

where  $\alpha = P_p/P_r(M_0)$ . We now go on to compute the hovering endurance using this equation.

Let the endurance be defined as the time  $t_e \equiv \xi_e \tau_r$  at which the entire mass of fuel  $M_f$  has been consumed and when the dimensionless mass is  $\mu(t_e) \equiv \mu_e = 1 - M_f/M_0$ . In the Appendix, we derive an approximate solution to eq. (15) that is linear in the small parameter  $\alpha$ :

$$\xi = 2(\mu^{-1/2} - 1) - \frac{\alpha}{2}(\mu^{-2} - 1) . \quad (16)$$

Figure 2 shows this approximation and the full numerical solution when  $\alpha = 0.01$  and  $0.1$ . As an example, if the fuel initially takes up 50% of the total mass of the helicopter, we see from eq. (16) and Fig. 2 that  $\xi_e \approx 0.7$ – $0.8$  when  $\alpha = 0.01$ – $0.1$ . In this case, the endurance is  $\sim \tau_r$ , which takes the numerical value

$$\tau_r \approx 160 \eta_f \eta_s \eta_t^{3/2} e^{-z/16} \left( \frac{A_r}{100 \text{ m}^2} \cdot \frac{10^3 \text{ kg}}{M_0} \right)^{1/2} \text{ hr} , \quad (17)$$

where we have used the density model in eq. (1),  $g = 9.8 \text{ m s}^{-2}$ , and  $\varepsilon_f = 50 \text{ MJ kg}^{-1}$ .

All that remains now is to somehow estimate the value of the combined efficiency parameter  $\eta_f \eta_s \eta_t^{3/2}$ . Most internal combustion engines have fuel efficiencies of  $\eta_f \approx 0.2$ – $0.3$  over a wide range of operating conditions, with the remainder of the chemical energy lost as heat and mechanical vibration. As for  $\eta_t$  and  $\eta_s$ , our expectation is that these parameters can be brought close to their optimal values—presumably  $\sim 1$ —by careful design of the fuselage and rotor system. A reasonable guess might be that  $\eta_f \eta_s \eta_t^{3/2} \approx 0.1$ , a value we will check in the next section.

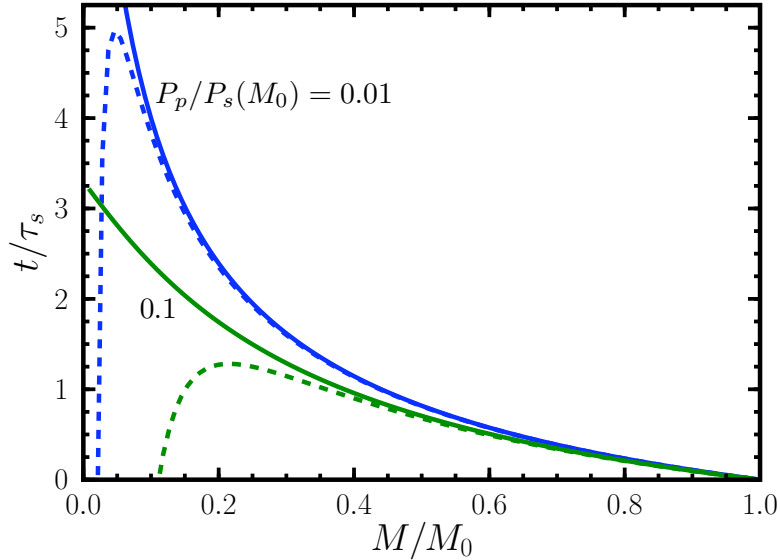


Figure 2: Flight time as a function of mass fraction. The top and bottom pairs of curves correspond to  $P_p/P_r(M_0) = 0.01$  (blue) and  $P_p/P_r(M_0) = 0.1$  (green), respectively. The solid curves are the results of numerical integration of the differential equation (eq. [15]), and the dashed curves are approximations linear in  $\alpha$  (eq. [16]).

## 4 The Hummingbird

Boeing's A-160 Hummingbird is listed<sup>1</sup> as having an empty mass (no fuel) of  $\approx 1100$  kg and a fueled take-off mass of  $\approx 2000$  kg without any payload, giving a fuel mass of  $\approx 900$  kg. Therefore, when there is no payload mass and all the power goes to the rotor, we have  $M_f/M_0 \approx 0.45$  and  $\alpha = 0$ , and so  $\xi_e \approx 0.7$  (eq. [16]). For the 11 m rotor diameter of the A-160, the endurance at a hover altitude of 5 km is

$$t_e = \xi_e \tau_r \approx 6 \left( \frac{\eta_f \eta_s \eta_t^{3/2}}{0.1} \right) \text{ hr} . \quad (18)$$

The quoted endurance of the A-160 at 5 km is 20 hr, which can be accommodated in our model if  $\eta_f \eta_s \eta_t^{3/2} \approx 0.3$ . It would seem that the A-160 must have a nearly optimal design in terms of its engine performance, fuselage and rotor design, and material choices. Figure 3 shows that the A-160 does indeed have a very streamlined appearance.

The A-160 is rated to carry up to a 500 kg payload. If we again assume  $\alpha = 0$  and increase the total initial mass to 2500 kg, our model predicts that the hover endurance is reduced by a factor of  $\approx 0.65$ , giving a time of  $\approx 13$  hr if the nominal endurance is 20 hr without any payload. An external payload, such as the one shown in Fig. 3 may compromise the aerodynamic performance of the helicopter and significantly decrease the combination  $\eta_s \eta_t^{3/2}$ , resulting in a shorter endurance.

## 5 Battery Powered Helicopters

Up to this point, our focus has been on helicopters powered by internal combustion engines. It is also of interest to consider helicopters powered by batteries. Such a platform would have no exhaust or engine

<sup>1</sup>See, e.g., <http://www.designation-systems.net/dusrm/app2/q-18.html> and [http://en.wikipedia.org/wiki/Boeing\\_A160\\_Hummingbird](http://en.wikipedia.org/wiki/Boeing_A160_Hummingbird).



Figure 3: A-160 Hummingbird. The appendage on the belly of the craft is the antenna for the FORESTER radar system.

noise, increasing the possibility for stealthy activities, and could be easily recharged electrically.

Current high-performance batteries<sup>2</sup> have energy densities of  $\lesssim 1 \text{ MJ kg}^{-1}$ , a small fraction of the energy density of combustible fuels. Let us denote the battery energy density by  $\varepsilon_b$ . For a given desired minimum flight time  $t_f$  and constant total helicopter mass  $M_0$ , what battery mass  $M_b$  is required? In this exercise, we will assume that all of the electrical power provided by the battery goes to the rotor and that the battery can be completely discharged. The minimum battery mass is then

$$M_{b,\min}(t_f) = \frac{P_r t_f}{\varepsilon_b} , \quad (19)$$

where  $P_r$  is the shaft power given by eq. (9). This is a soft limit, since its value is determined as much by the adjustable flight time as by the limitations of the battery. Numerically, we have

$$M_{b,\min}(t_f) \approx 320 (\eta_s \eta_t^{3/2})^{-1} e^{z/16} \left( \frac{t_f}{1 \text{ hr}} \right) \left( \frac{1 \text{ MJ kg}^{-1}}{\varepsilon_b} \right) \left( \frac{M}{10^3 \text{ kg}} \right)^{3/2} \left( \frac{100 \text{ m}^2}{A_r} \right)^{1/2} \text{ kg} . \quad (20)$$

If all of the variables above are set to the reference values shown, the battery mass must be more than 30% of the total helicopter mass. We will explore the scaling of this mass fraction with helicopter size in § 6.

In addition to a fixed total energy, batteries are limited in how much power they can provide. Modern batteries can supply a maximum power per unit mass of  $\lesssim 1 \text{ kW kg}^{-1}$ . Denote the maximum power density by  $\kappa_b$ . This limit provides a hard minimum on the battery mass given the power requirements of the helicopter rotor:

$$M_b > \frac{P_r}{\kappa_b} . \quad (21)$$

A battery mass equal to the minimum value,  $P_r/\kappa_b$ , corresponds to a minimum flight endurance (see eq. [19]),

$$t_{e,\min} = \frac{\varepsilon_b}{\kappa_b} = 10^3 \left( \frac{\varepsilon_b}{1 \text{ MJ kg}^{-1}} \cdot \frac{1 \text{ kW kg}^{-1}}{\kappa_b} \right) \text{ s} . \quad (22)$$

A decrease in  $\kappa_b$  will increase the minimum endurance, but at the expense of an increase in the minimum battery mass.

<sup>2</sup>For information on lithium-ion batteries, see, e.g., [http://en.wikipedia.org/wiki/Lithium-ion\\_battery#cite\\_note-2](http://en.wikipedia.org/wiki/Lithium-ion_battery#cite_note-2) .

While the minimum endurance is a function of the battery microphysics, the theoretical maximum endurance arises when the battery mass equals the entire mass of the helicopter:

$$t_{e,\max} = \frac{\varepsilon_b M_0}{P_r} . \quad (23)$$

Comparing this time with eq. (13) for  $\tau_r$ , the approximate endurance of a helicopter powered by an internal combustion engine, we find

$$\frac{t_{e,\max}}{\tau_r} = \frac{\varepsilon_b}{\eta_f \varepsilon_f} \approx 0.02 \eta_f^{-1} \left( \frac{\varepsilon_b}{1 \text{ MJ kg}^{-1}} \cdot \frac{50 \text{ MJ kg}^{-1}}{\varepsilon_f} \right) . \quad (24)$$

If endurance is the primary concern, internal combustion engines are certainly favored over batteries. However, battery-powered electrical motors are apparently easier to scale to very small sizes than are internal combustion engines. Batteries are the logical choice for helicopters with characteristic lengths much less than 1 m.

## 6 Similarity Scaling

Here we extend an idea first touched on below eq. (5), where we attempted to use a scaling argument to estimate  $\tan \theta$ . In that exercise, we wrote the mass as  $M = \bar{\rho} A_d L$ , the product of a mean fuselage density, fuselage cross sectional area, and characteristic fuselage length. Imagine now that we build a new helicopter with a length of  $2L$  by scaling up every component of the original model using the same materials and while exactly preserving all the relative geometric proportions. With these assumptions, the mean density is unchanged, the fuselage cross sectional area becomes  $4A_d$ , and the total mass of the new helicopter is  $8M$ . By assuming geometric similarity, we obtain the scaling relation  $M \propto L^3$ , where  $L$  can be any length associated with the helicopter.

Two helicopter size-related parameters appear repeatedly in the previous sections: the helicopter mass  $M$  and the rotor area  $A_r$ . According to the similarity assumption,  $M \propto L^3$  and  $A_r \propto L^2$ . It follows that the air flow speed and rotor power (eqs. [7] and [9]) scale as

$$u_r \propto \left( \frac{M}{A_r} \right)^{1/2} \propto L^{1/2} , \quad (25)$$

and

$$P_r \propto \left( \frac{M^3}{A_r} \right)^{1/2} \propto L^{7/2} . \quad (26)$$

In the case of an internal combustion engine and a fixed fuel mass fraction  $M_f/M_0$ , the hover endurance is a constant times  $\tau_r$  in eq. (13), and thus the endurance scales as

$$t_e \propto \left( \frac{A_r}{M_0} \right)^{1/2} \propto L^{-1/2} . \quad (27)$$

Whether or not the above exponents are exactly matched in real systems, it is at least plausible that  $P_r$  is a strong increasing function of  $L$  and that  $t_e$  has a relatively weak decreasing trend. If endurance is the key requirement, we may gain by a factor of  $\approx 3$  in moving from helicopters with 10 m rotor diameters to those with lengths of  $\approx 1$  m. Of course, this assumes that the size and power of useful payloads can be reduced accordingly.

We showed in § 5 that the minimum battery mass is  $\propto P_r$ , whether the constraint is a given flight duration (eq. [19]) or the maximum power density of the battery (eq. [21]). Therefore,

$$M_{b,\min} \propto P_r \propto L^{7/2} , \quad (28)$$

and

$$\frac{M_{b,\min}}{M_0} \propto L^{1/2} . \quad (29)$$

Since the mass fraction  $M_{b,\min}/M_0$  increases with  $L$ , there is a maximum  $L$ ,  $L_{\max}$ , at which  $M_{b,\min}/M_0 = 1$ . The helicopter cannot be made larger than  $L_{\max}$ ; the value of this maximum scale depends on design parameters, such  $\eta_s$ ,  $\eta_t$ , and  $\kappa_b$ . Now suppose we adopt a fixed battery mass fraction  $M_b/M_0$  for all  $L$ , making sure that  $M_b > M_{b,\min}$  at some  $L_0$ . As we decrease  $L$  below  $L_0$ , we can be assured that the battery mass is always greater than  $M_{b,\min}$  according to eq. (29).

## A Approximate Solution of the Endurance Equation

We begin with eq. (15):

$$\frac{d\mu}{d\xi} + \mu^{3/2} = -\alpha , \quad (A1)$$

with the initial condition  $\mu(0) = 1$ . When  $\alpha = 0$ , the solution to eq. (15) is immediate, and we see that  $\xi$  is related linearly to  $\mu^{-1/2}$ . Therefore, a sound alternative choice for the dependent variable is  $\zeta = \mu^{-1/2}$ , which yields

$$2\frac{d\zeta}{d\xi} = 1 + \alpha\zeta^3 . \quad (A2)$$

This equation has the formal solution

$$\xi = 2 \int_1^\zeta dv (1 + \alpha v^3)^{-1} . \quad (A3)$$

This integral can be evaluated in terms of elementary functions, but the result is not terribly illuminating; for our purposes, a simple approximation is sufficient. After expanding the integrand in a Taylor series, the exact solution to eq. (A2) is

$$\begin{aligned} \xi &= 2 \int_1^\zeta dv \sum_{n=0} (-1)^n \alpha^n v^{3n} = 2 \sum_{n=0} \frac{(-1)^n \alpha^n}{3n+1} (\zeta^{3n+1} - 1) \\ &= 2 \left[ (\zeta - 1) - \frac{\alpha}{4} (\zeta^4 - 1) + \dots \right] . \end{aligned} \quad (A4)$$

One should be concerned about the rate of convergence of the series when  $\zeta$  is relatively large. As long as  $\alpha\zeta^3 \ll 1$ , we are justified in keeping only the first two terms in the series. As a function of  $\mu$ , the  $\mathcal{O}(\alpha)$  approximation is

$$\xi = 2(\mu^{-1/2} - 1) - \frac{\alpha}{2}(\mu^{-2} - 1) . \quad (A5)$$

The range of validity of this approximation is demonstrated in Fig. 2.